

Question Paper

Exam Date & Time: 27-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

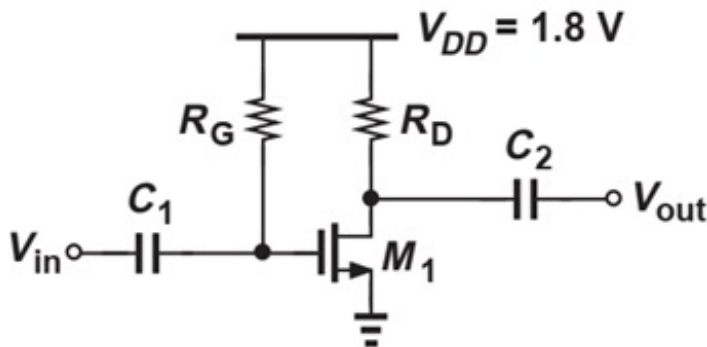
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ECE 2121/ECE_2121 ANALOG ELECTRONIC CIRCUITS
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

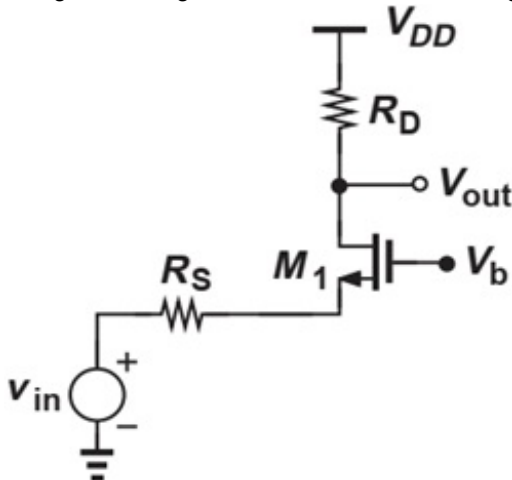
Answer all the questions.

- 1A) The circuit shown below must be designed for a voltage gain of -5 with a power budget of 2 mW. If $R_{D1D} = 1\text{ V}$, determine the required value of W/L . Assume the capacitors are very large, $R_G = 10\text{ K}\Omega$, $\mu_n C_{ox} = 200\text{ }\mu\text{A/V}^2$, $\lambda=0$, and $V_{TH}=0.4\text{ V}$. (4)



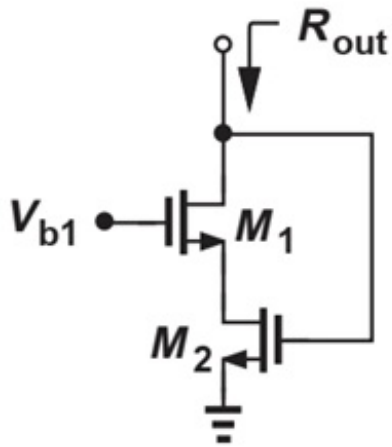
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 1B) Using a small signal model, calculate the voltage gain for the circuit shown below. Consider $\lambda=0$. (3)



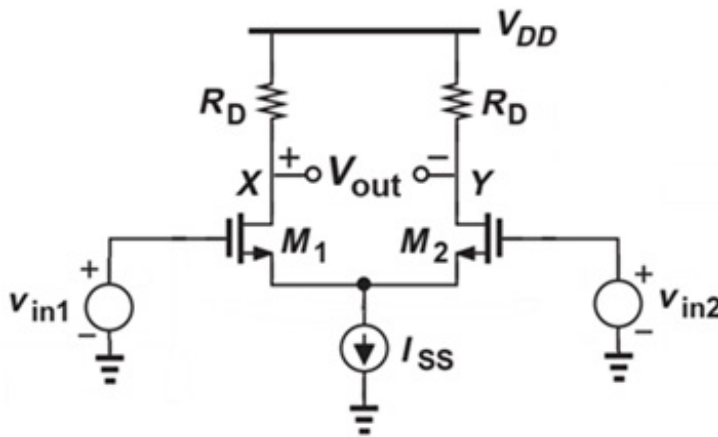
Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1C) Using the small signal model, compute the output resistance of the circuits depicted below. Assume all the transistors operate in saturation. Consider $\lambda \neq 0$. (3)



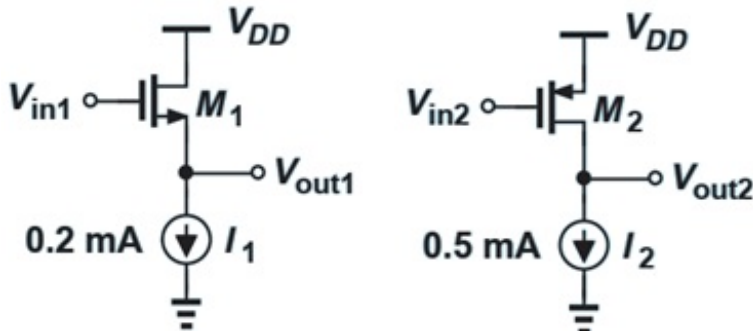
Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Design the MOS differential amplifier shown below to operate at an overdrive voltage of 0.25 V and to provide a transconductance of 1 mA/V. Consider $V_{TH}=0.5V$ and $\mu_n C_{ox}=100 \mu A/V^2$. (4)



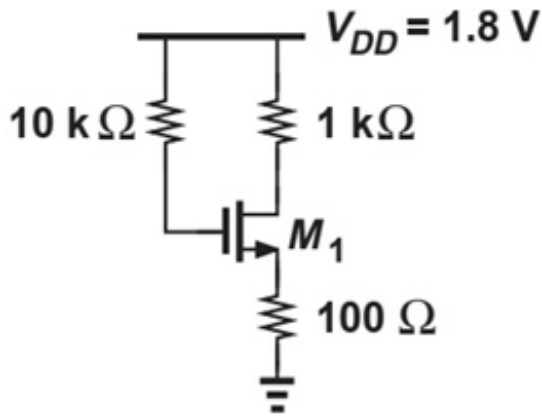
Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2B) An integrated circuit employs the source follower and the common-source stage, as shown in the figure below. Design a current mirror that produces I_1 and I_2 from a 0.3mA reference. Assume $\lambda=0$. (3)



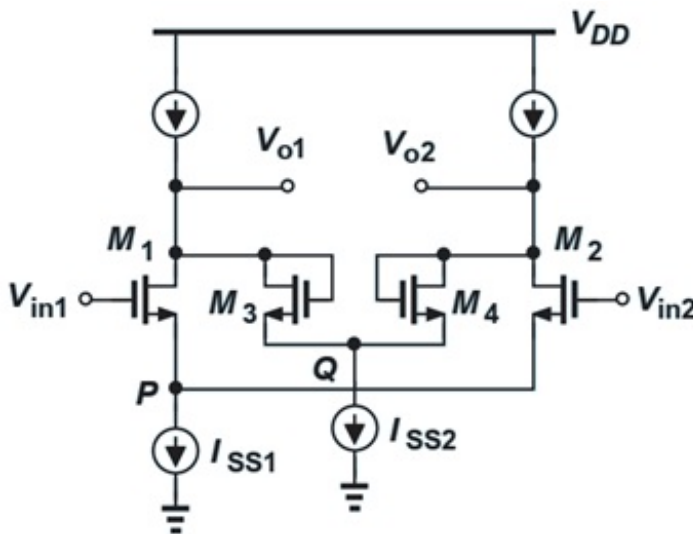
Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) Consider the circuit shown below. Calculate the maximum transconductance that M_1 can provide (without going into the triode region). Assume $\mu_n C_{ox} = 200 \mu A/V^2$, $\lambda=0$, and $V_{TH}=0.4V$. (3)



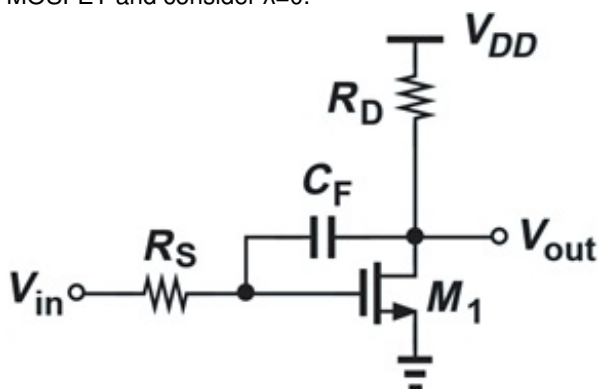
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Determine the expression of single-ended differential voltage gain for the circuit shown below. Assume perfect symmetry and $\lambda \neq 0$. (4)



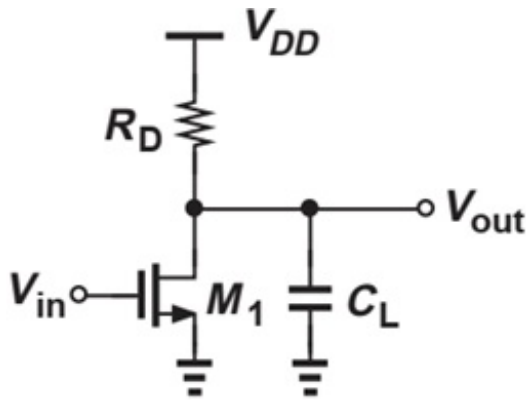
Course Outcome: "CO2" / Cognitive Level: "Application"

- 3B) For the amplifier shown, determine the pole frequencies. Ignore the parasitic capacitance of the MOSFET and consider $\lambda=0$. (3)



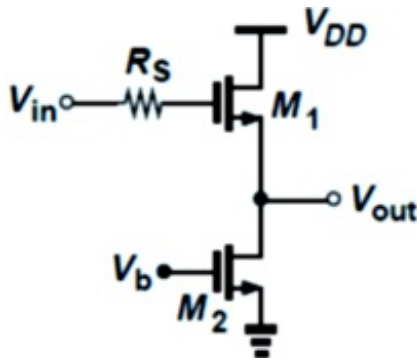
Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) In the circuit shown below, $R_D = 1\text{ k}\Omega$ and $C_L = 1\text{ pF}$. Neglecting channel-length modulation and other capacitances. Determine the frequency at which the gain falls by 10%. (3)



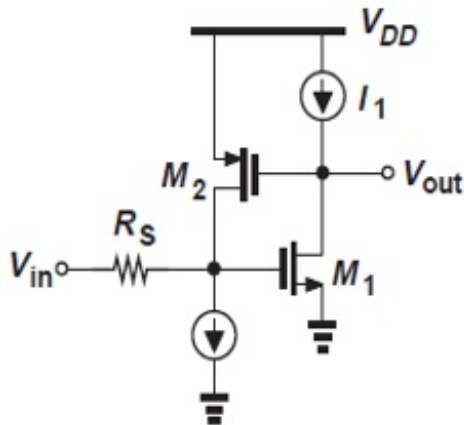
Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Identify all the capacitance and reduce the source follower circuit shown to compute the poles. Assume $\lambda=0$. (4)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) Determine the polarity of the feedback of the circuit and justify the same. (3)

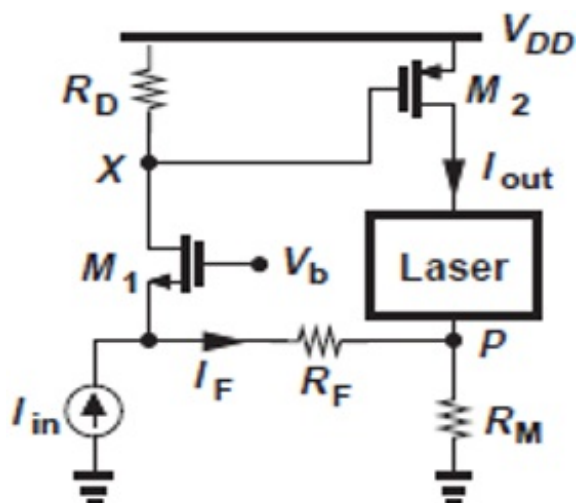


Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4C) Explain the working principle of the Colpitts Oscillator with a neat circuit diagram. Also, obtain an expression for frequency of oscillation and find the frequency of oscillation for $L1 = 2\text{mH}$ and $C1 = C2 = 10\mu\text{F}$. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5A) Analyze the circuit shown where the output current delivered to a laser is regulated by negative feedback taking loading effect of feedback into account. Compute the closed-loop gain and I/O impedances, considering $\lambda=0$ and model the laser as resistance R_L . (4)



Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Explain the working principle of Class - B power amplifier with a neat circuit diagram. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5C) Explain the working principle of RC phase shift oscillator with a neat circuit diagram and relevant expressions. Also, find the frequency of oscillation for capacitance of $0.01\mu\text{F}$ and resistance of $10\text{K}\Omega$. (3)

Course Outcome: "CO4" / Cognitive Level: "Understand"

-----End-----

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_2121:ANALOG ELECTRONIC CIRCUITS

#	Course Outcome	Description	Marks
1	1	CO 1	6
2	2	CO 2	18
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	6
Total			50

* **NA** - CO Not Selected

Event Name	Subject
------------	---------

058-END SEMSTER-ODD-2025 ECE_2121:ANALOG ELECTRONIC CIRCUITS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	8	27(54%)
4	Analysis	2	8(16%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)